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Exercise intensity is an important determinant of changes in muscle monocarboxylate transporter abundance in active men

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Dynamic cerebral autoregulation

	Baseline		LPS	
	Normotension	Hypertension	Normotension	Hypertension
MAPsp (mmHg ² Hz ⁻¹)	2.9 (2.6-3.6)	7.9 (2.1-8.9)	1.5 (0.9-4.9)	5.2 (2.1-6.9)
MCAvsp (cm sec ⁻² Hz ⁻¹)	5.6 (4.3-6.2)	7.3* (4.8-8.8)	2.0† (1.7-4.6)	4.5 (2.6-7.1)
Gain (cm mmHg ⁻¹ sec ⁻¹)	1.17 (1.12-1.35)	0.89* (0.78-0.97)	0.86† (0.76-0.89)	0.81 (0.77-0.92)
Phase (radians)	0.71 (0.67-0.93)	0.93 (0.91-1.10)	1.14† (1.10-1.35)	1.48† (1.21-1.62)
Coherence (units)	0.83 (0.82-0.88)	0.83 (0.69-0.87)	0.66† (0.57-0.79)	0.67 (0.66-0.83)

LPS was infused at a total dose of 2 ng kg⁻¹. MAPsp: MAP spectral power; MCAvsp: MCAv spectral power. Data are presented as median (interquartile range). Different from normotension in the same condition (i.e. baseline or LPS), * $p < 0.05$. Different from the corresponding intervention (i.e. normotension or hypertension) at baseline, † $p < 0.05$.

Zhang et al. *J Physiol* 587, 2567–2577, 2009.

Sharshar et al. *Intensive Care Med* 33, 798–806, 2007.

Zhang et al. *Am J Physiol* 274, H233–H241, 1998.

Berg et al. *Crit Care* 17, R238, 2013.

Where applicable, the authors confirm that the experiments described here conform with The Physiological Society ethical requirements.

C40

Hypoxia suppresses release of brain derived neurotrophic factor from the human brain

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Background and Aims: Brain derived neurotrophic factor (BDNF) is a key protein involved in neurogenesis, influencing the regulation, growth and survival of neurons, learning and memory (Chen *et al.*, 2013). Its systemic concentration has been shown to rise in response to acute hypoxia (Hubold *et al.*, 2009) though to what extent the human brain contributes remains to be established. Thus, in the present study, we measured the jugular venous to radial arterial concentration difference (JV-RA_{DIFF}) to document, for the first time, the trans-cerebral exchange kinetics of BDNF during hypoxia. We also chose to measure nitrite (NO₂⁻) exchange since cellular evidence suggests that BDNF formation may be regulated by nitric oxide (NO) (Xiong *et al.*, 2009).

Methods: Ten healthy males were examined in normoxia and following 12h passive exposure to poikilocapnic normobaric hypoxia (~13% O₂). Blood samples were obtained simultaneously from the RA and internal JV. Plasma was assayed for NO₂⁻ using ozone-based chemiluminescence and platelet-poor BDNF via enzyme-linked immunosorbent assay (R&D Systems, Minneapolis, MN, USA). Global cerebral blood flow (CBF) was determined in the desaturation mode using inhaled nitrous oxide (5%) as the tracer (Kety and Schmidt, 1945). Cerebral plasma flow (CPF) was calculated as CBF x (1-haematocrit) and trans-cerebral exchange calculated as CPF x JV-RA_{DIFF}. Following confirmation of distribution normality using Shapiro-Wilk W tests JV-RA_{DIFF} data were analysed with a two factor repeated measures ANOVA and post-hoc Bonferroni-corrected paired samples t-tests. Exchange data were analysed using a paired samples t-test expressed as mean ± standard deviation (SD).

Results: Hypoxia was associated with a decrease in arterial oxygen content (8.5 ± 0.4 to 7.1 ± 0.4 mmol/L, $P < 0.05$) whereas no changes were observed in CPF (51 ± 9 to 57 ± 7 mL/100g/min, $P > 0.05$) or platelet count (190 ± 51 vs. 177 ± 76 x 10⁹/L, $P > 0.05$). A net cerebral output of BDNF (169 ± 300 pg/min/g) and uptake of NO₂⁻ (-126 ± 94 nmol/min/g) was apparent at baseline during normoxia. Hypoxia was shown to suppress ($P < 0.05$ vs. normoxia) both BDNF output (25 ± 94 pg/min/g) and NO₂⁻ uptake (-16 ± 47 nmol/min/g) due to a reduction in JV outflow and RA inflow respectively.

Conclusions: The present findings fail to confirm the systemic rise in BDNF previously observed during hypoxia (Hubold *et al.*, 2009) and further exclude the human brain as a contributory source. To the contrary, the cerebral output of BDNF was shown to decrease during hypoxia and the concomitant reduction in NO₂⁻ uptake suggests that its cerebral formation may be regulated by NO consumption.

Chen *et al.* (2013). *Biomed Reports* 1, 167-176.

Hubold *et al.* (2009). *J Neural Transm* 116, 1221-1225.

Kety SS and Schmidt CF (1945). *Am J Physiol* 143, 53-66.

Xiong *et al.* (1999). *Eur J Neurosci* 11, 1567-1576.

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C41

Exercise intensity is an important determinant of changes in muscle monocarboxylate transporter abundance in active men

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High-intensity interval training (HIIT) has often been found to enhance muscle pH regulation through increases in muscle buffer capacity (1) and H⁺ transport proteins (2). During exercise the largest portion of H⁺ regulation comes from the monocarboxylate transporters (MCT1 and MCT4) (3). The signalling factors responsible for upregulation of H⁺ transport proteins are uncertain (4), although H⁺ accumulation or H⁺ efflux may be important stimuli (5). Maximising high non-mitochondrial ATP turnover (H⁺ accumulation) and exercise duration (H⁺ flux) may prove to be an optimal stimulus. We hypothesised that more intervals performed just above the lactate threshold (LT) would produce greater changes in MCT content than fewer intervals closer to W_{peak}. We also investigated if six weeks of training cessation would be sufficient for MCT content to return to baseline.

Using a two-group parallel design, 16 men (23 ± 5 y, mean ± SD) completed 4 weeks of work-matched HIIT at either 20% (LO, n=8) or 94% (HI, n=8) of the difference between LT and W_{peak}. VO_{2peak} was measured pre- (0wk) and post-training (+4wk). Repeated-sprint ability (RSA) was assessed at 0wk, +4wk, and after 6 weeks training cessation (+10wk). Western blotting was used to measure MCT1 and MCT4 abundance in muscle samples taken at 0wk, +2wk, +4wk, and +10wk. Data were analysed using a two-way mixed design ANOVA. Cohen's *d* effect sizes (ES) are presented as (ES; 90% confidence interval) of the difference scores.

MCT1 abundance (Fig.1) at +4wk increased more for LO than HI (ES=0.47; -1.35 to 2.28). MCT4 changes followed different time-courses for the two groups (P=0.007). HI showed an improvement in MCT4 at +2wk but had lost this improvement

at +4wk, whereas LO improved only at +4wk (Fig.2). MCT abundance returned to baseline for both groups at +10wk. LT improved in response to training more for LO than HI ($14 \pm 13\%$ vs $2 \pm 9\%$; $ES=0.36$; -0.03 to 0.75). The modest increases in W_{peak} showed no difference between groups ($3 \pm 4\%$ and $3 \pm 6\%$ for LO and HI respectively). There was little change in total work ($2 \pm 7\%$ and $-2 \pm 6\%$ for LO and HI respectively) or fatigue decrement during the RSA test. There were no changes in VO_{2peak} for either group.

The large but variable increases in MCT1 abundance for both groups provides unclear evidence as to whether prolonged H^+ flux is a key factor for upregulation. Some of the equivocation in MCT4 training adaptation may be explained by differing time-courses of adaptation depending on the intensity of HIIT. The reduction in MCT4 at +4wk and lack of improvement in LT compared to LO hint that 4 weeks of HI training may have caused overreaching. Finally, 6 weeks of training cessation resulted in MCT protein abundance returning to baseline for both groups.

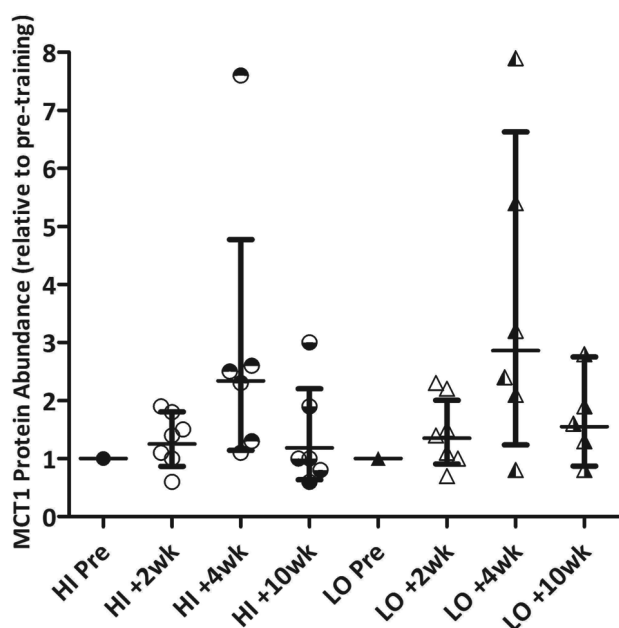


Figure 1. MCT1 protein abundance for the high-intensity (HI) and low-intensity (LO) groups pre-training, after 2 and 4 weeks of training, and after a further 6 weeks of training cessation. Data are expressed relative to a calibration curve of internal standards and normalised to pre-training values. Individual data points are plotted along with geometric mean and 95% confidence intervals. CV of duplicates 9%.

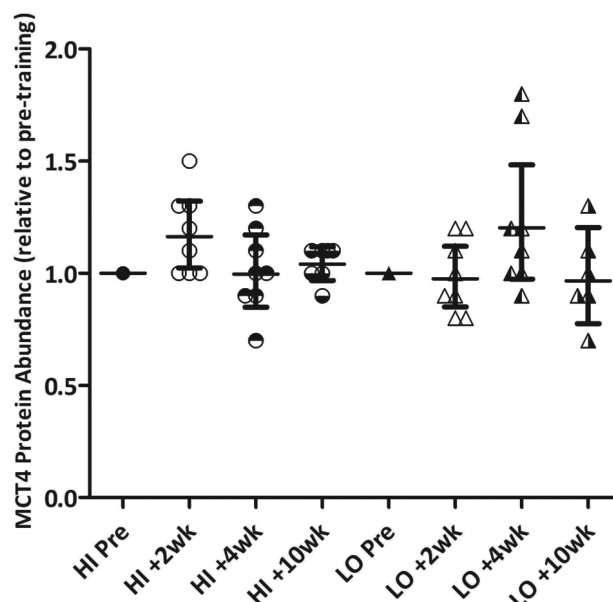


Figure 2. MCT4 protein abundance for the high-intensity (HI) and low-intensity (LO) groups pre-training, after 2 and 4 weeks of training, and after a further 6 weeks of training cessation. Data are expressed relative to a calibration curve of internal standards and normalised to pre-training values. Individual data points are plotted along with geometric mean and 95% confidence intervals. CV of duplicates 6%.

Edge J, et al. Eur J Appl Physiol. 2006;96(1):97-105.

Burgomaster KA, et al. Am J Physiol Regul Integr Comp Physiol. 2007;292(5):R1970-6.

Juel C, et al. J Physiol. 1999;517(3):633-42.

Thomas C, et al. Am J Physiol Regul Integr Comp Physiol. 2012;302(1):R1-R14.

Weston AR, et al. Acta Physiol Scand. 1996;157(2):211-6.

Supported in part by an Exercise & Sports Science Australia Applied Sports Science Research Grant.

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C42

Renal vasoconstriction during isometric calf exercise is augmented in older, healthy humans

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Cardiovascular control during exercise is regulated in part by reflex feedback from muscle afferent nerve fibres (McCloskey and Mitchell, 1972). Isometric handgrip exercise causes renal vasoconstriction in humans (Middlekauff et al., 1997) and this effect is augmented in older, healthy humans (Momen et al., 2004). However, it is unknown whether healthy ageing augments renal vasoconstriction during isometric calf exercise. Exaggerated renal vasoconstriction occurs during mild, plantarflexion exercise in peripheral arterial disease patients (Drew et al., 2013) so elucidating the effect of ageing alone on renal responses during leg exercise is of clinical relevance. Therefore, this study examined renal vascular responses during isometric, calf exercise in healthy young and older humans. Twelve young, healthy (Young; 6 men; mean $\pm$ SEM 25 ± 1 y) and ten older, healthy (Older; 4 men; 62 ± 1 y) subjects rested for 5